Vinayak Paroonon Kooloth

Agentic AI Engineer | LLM Orchestration & Automation | Multi-Agent Systems | EU AI Compliance

Oulu, Finland | Open to relocation across EU

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Profile

AI Engineer specialising in agentic AI systems, multi-agent orchestration, LLM automation pipelines, and backend AI integration. Hands-on experience architecting multi-agent clinical AI systems, autonomous internal knowledge agents, evidence-grounded research agents, agentic retrieval pipelines, and FastAPI ML microservices. Proficient with agentic coding tools including Claude Code and GitHub Copilot/Codex for accelerating AI pipeline development. Experienced building hybrid RAG evaluation pipelines with automated quality gates, citation-correctness metrics, and observability tracing. Strong background in EU AI Act risk classification, GDPR data-minimisation architecture, and MDR compliance.

Education

Hof University of Applied Sciences — Hof, Germany
Master of Science in Artificial Intelligence & Robotics — *Mar. 2024 – Present*

CMS College of Engineering and Technology — India
Bachelor of Engineering in Computer Science and Engineering — *Aug. 2019 – Jul. 2023*

Experience

Monidor Oy — Oulu, Finland
AI & Full-Stack Engineer — Medical Systems — *Jan. 2026 – Present*

- Architected a multi-agent clinical AI system with specialised agents for intent classification (XLM-RoBERTa), patient-history retrieval, and LLM-based response generation, coordinated through a central agentic orchestration layer.
- Designed an agentic dispatcher that autonomously routes clinical queries to the optimal retrieval strategy — standard RAG, temporal retrieval, multi-hop retrieval, or hybrid dense–sparse search — based on query intent, replacing a single monolithic pipeline.
- Evaluated retrieval strategies across relevance, latency, and answer quality dimensions, feeding results back into the routing agent to enable adaptive, self-improving agent behaviour over time.
- Built a Node.js and PostgreSQL reporting backend for patient vital-sign data (SpO_2, $EtCO_2$, capnography waveforms), including robust data-validation logic for missing sensor readings and device error states.
- Leveraged Claude Code and CLI-based agentic coding workflows to scaffold agent modules, automate prompt iteration, and accelerate clinical AI pipeline development.
- Drove EU AI Act, GDPR, and MDR compliance architecture: designed audit-logging pipelines, enforced data-minimisation at ingestion, implemented transparency and explainability layers, and conducted AI risk classification for a high-risk medical AI system.

Fricke und Mallah Microwave Technology GmbH — Peine, Germany
Working Student — Agentic AI & Automation — *Jun. 2025 – Sep. 2025*

- Built an autonomous internal AI agent using LangChain and Mistral-7B that uses CRM lookup, ERP project search, and documentation retrieval as agent tools, enabling end-to-end automation of reporting, follow-up, and knowledge lookup workflows.
- Engineered an agentic automation pipeline with FAISS-based vector retrieval for email triage and internal knowledge search, reducing manual processing effort by over 60% in internal workflow testing.
- Prepared CRM, ERP, and email corpora for agent-ready embedding search: cleaning, normalising, chunking, metadata tagging, and building FAISS vector indexes for tool-use retrieval.
- Used agentic coding tools (GitHub Copilot/Codex) to accelerate prompt engineering, pipeline scaffolding, and multi-lingual output refinement.
- Benchmarked Mistral-7B and Llama 3 for autonomous task completion — automated reporting, multilingual summarisation, and knowledge extraction — and presented model selection recommendations to management.

Technische Universität Chemnitz — Chemnitz, Germany
Student Assistant — Surgical Robotics Simulation — *Mar. 2025 – Oct. 2025*

- Built perception-to-action agent demos for surgical lamp and robot interaction using MediaPipe, OpenCV, ROS2, and Gazebo, implementing a full sense–decide–act loop for gesture-based control.
- Prototyped a multi-joint robotic arm simulation in Unity and RViz for an EU-funded research project, creating a reusable model for cross-lab experiment replication and clinical partner collaboration.

Hof University of Applied Sciences — Hof, Germany
Student Researcher — Applied Machine Learning — *Oct. 2024 – Mar. 2025*

- Developed an explainable phishing-detection pipeline using DistilBERT and SHAP to classify suspicious text and surface model predictions for end users.
- Deployed the model as a FastAPI microservice with real-time inference and used MLflow to track accuracy, loss, and inference latency across training runs.

Bachelor's Thesis — India
Banana Leaf Disease Detection Using Hybrid Deep Learning Model — *2023*

- Developed a hybrid CNN–LSTM model for banana leaf disease classification, achieving 94.5% test accuracy on Sigatoka and Xanthomonas disease detection.
- Curated and augmented a dataset from the Mendeley repository and compared CNN and CNN–LSTM models using precision, recall, latency, and training-epoch analysis.

Selected AI Projects

Evidence-Grounded Multi-Agent Research System

- Designed a graph-based multi-agent workflow spanning planner, query expansion, multi-source discovery, source-quality scoring, document ingestion, hybrid retrieval, evidence synthesis, critic review, and automated evaluation agents.
- Implemented a hybrid RAG pipeline combining BM25, vector retrieval, metadata filtering, and reranking (Cohere Rerank, bge-reranker) to maximise evidence relevance and eliminate hallucinated claims.
- Developed a source-quality scoring engine using authority, recency, relevance, evidence depth, citation support, primary-source ratio, and bias-risk features across papers, official docs, case studies, forums, and code repositories.
- Built automated evaluation pipelines measuring faithfulness, answer relevancy, context precision/recall, citation correctness, source diversity, and hallucination rate using RAGAS and custom evaluators.
- Created CI-style quality gates blocking final reports when faithfulness, citation correctness, source diversity, or unsupported-claim thresholds fail, with automated retry and re-retrieval paths.
- Integrated end-to-end observability with structured agent traces, token and cost tracking, latency monitoring, tool-failure logging, and an evaluation dashboard via LangSmith and Arize Phoenix.
- Generated structured research reports with claim-level evidence tables, confidence scores, contradiction analysis, source maps, and traceable citations across heterogeneous source types.

Technical Skills

Agentic AI & Automation: Multi-agent orchestration, LangGraph, LangChain Agents, tool-use & function calling, agentic dispatcher design, graph-based agent workflows, quality gates, Claude Code, GitHub Copilot/Codex, CLI-based agentic coding, autonomous pipeline automation

LLM & RAG: LangChain, LlamaIndex, FAISS, Qdrant, pgvector, hybrid retrieval (BM25 + vector), reranking (Cohere Rerank, bge-reranker), RAGAS, DeepEval, Phoenix Evals, LangSmith, Arize Phoenix, citation-correctness evaluation, source-quality scoring, prompt engineering, vector search, adaptive retrieval routing

AI/ML: PyTorch, TensorFlow, Scikit-learn, Transformers, DistilBERT, XLM-RoBERTa, CNNs, LSTMs, SHAP, MLflow

EU AI Compliance: EU AI Act (risk classification, audit logging, transparency), GDPR data-minimisation & privacy-by-design, MDR considerations for clinical AI systems

Backend & Data: FastAPI, REST APIs, Node.js, PostgreSQL, OpenSearch, SQLite, Docker, GitHub Actions, CI/CD, server-rendered reports

Cloud & Data Tools: AWS S3, AWS Lambda, AWS SageMaker, Azure, Power BI, sentence-transformer embeddings

Computer Vision & Robotics: OpenCV, MediaPipe, YOLOv7, ROS2, Gazebo, Unity3D, RViz, robotic simulation

Programming: Python, TypeScript, JavaScript, Node.js, C++, SQL, Bash

Languages

English: C1 — Professional proficiency **German:** B1 — Working proficiency **Finnish:** A1 — Beginner
Hindi / Malayalam: Native / Fluent

Interests

Agentic AI Systems, Multi-Agent Orchestration, LLM Automation, Multilingual NLP, Machine Learning, Computer Vision, Robotics Simulation, Cloud Automation, Hiking, Science Fiction